

Sumal Dasoju

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Professional Summary

Dynamic Data Engineer | Analyst | Scientist with hands-on experience across cloud data engineering, machine learning, analytics engineering, and large-scale ETL automation. Skilled in building end-to-end pipelines, developing ML models, transforming raw datasets into actionable insights, and enabling real-time reporting for enterprise teams. Proficient in Python, SQL, Spark, Airflow, AWS, and Power BI/Tableau with a strong focus on data quality, performance optimization, and scalable architecture. Known for delivering reliable, high-impact data solutions that directly support business strategy and operational excellence.

Technical Skills

Programming & Scripting: Python (Pandas, NumPy, PySpark), SQL (Advanced), R, Java, Shell Scripting, JSON, YAML

Data Engineering & Big Data: Apache Airflow, Spark (Batch + Streaming), Hadoop, Kafka, Talend, Informatica, AWS Glue, dbt, Apache Flink, Delta Lake, Lake Formation

Cloud Platforms: AWS (S3, Redshift, Glue, Lambda, Athena, EMR, Kinesis), Azure (ADLS, Synapse), GCP (BigQuery, Dataflow)

Databases & Storage: PostgreSQL, MySQL, Snowflake, MongoDB, Cassandra, DynamoDB, SQL Server, Redis, Parquet/ORC

Machine Learning & Analytics: Scikit-learn, XGBoost, LightGBM, Time-Series Forecasting, Feature Engineering, Model Deployment, MLflow, Jupyter, statistical analysis

Visualization & BI Tools: Power BI, Tableau, Looker, Apache Superset, matplotlib, seaborn

DevOps, Automation & CI/CD: Terraform, Docker, Git/GitHub, GitLab CI/CD, Jenkins, Kubernetes, CloudFormation, Observability (CloudWatch, Grafana)

Integration & APIs: REST APIs, SOAP, OAuth 2.0, CDC Pipelines, Streaming APIs, ETL/ELT Workflows, Event-Driven Architecture

Data Governance & Quality: Great Expectations, Data Lineage, Metadata Management, Data Catalogs, Quality Scorecards, PII/PHI Compliance, Security Controls

Professional Experience

Senior Data Engineer

- Drove modernization of legacy ETL systems into cloud-native architecture, improving reliability and scalability for long-running jobs.
- Implemented comprehensive data quality scorecards and validation rules achieving **99.9% accuracy** across mission-critical datasets.
- Built reusable Python libraries for logging, monitoring, and transformation logic, standardizing engineering workflows across teams.
- Automated data governance workflows including lineage, cataloging, and PII compliance using Glue, Atlas, and custom pipelines.
- Designed batch + streaming hybrid architecture supporting both historical and near-real-time decision-making.

Data Engineer

Capital One | Plano, TX | Jan 2024 – Present

- Engineered high-performance ETL pipelines using Airflow, Glue, and Talend, increasing processing efficiency by **60%**.
- Architected AWS-based data lake (S3, Glue, Redshift Spectrum) serving petabyte-scale data workloads across finance & risk teams.
- Developed Spark-based batch & streaming pipelines reducing data latency from hours to **under 10 minutes**.
- Implemented enterprise ingestion patterns using Kafka, Kinesis, and REST microservices to unify 25+ source systems.
- Optimized Redshift schema design, distribution keys, and sort strategies improving query performance by **45%**.
- Reduced AWS compute/storage spend by **35%** via right-sizing, partitioning, compression, and lifecycle policies.
- Built automated data quality frameworks using PyDeequ and custom validation rules ensuring **99.8% data accuracy**.
- Implemented CI/CD pipelines for data infrastructure using Terraform, GitHub Actions, and Docker.

- Designed secure IAM roles, Lake Formation permissions, and encryption-at-rest/in-transit ensuring bank-level compliance.
- Collaborated with ML teams to operationalize model training pipelines and deploy features to real-time systems.

Data Analyst / Data Scientist

Fidelity | Westlake, TX | Jan 2023 – May 2024

- Built and deployed predictive ML models (classification, regression, time series) improving forecast accuracy by **40%** and reducing variance in key financial KPIs.
- Analyzed 10M+ row datasets using Python, SQL, Pandas and delivered insights that saved **\$2.1M** in operational inefficiencies.
- Designed complex analytical data models supporting advisor insights, portfolio performance, and compliance metrics.
- Developed 20+ enterprise-grade Power BI dashboards used daily by **300+ business stakeholders**, enabling real-time decision-making.
- Automated end-to-end ETL & reporting pipelines using Airflow + Python, reducing manual workload by **70%**.
- Built scalable ingestion workflows integrating REST APIs, S3 datasets, market feeds, and internal warehouse sources.
- Implemented advanced feature engineering and model optimization, improving model recall, precision, and ROC scores by **25–30%**.
- Designed SQL-based analytical layers (views, materialized views, aggregation tables) reducing dashboard latency by **55%**.
- Conducted exploratory data analysis (EDA) and statistical tests (ANOVA, chi-square, correlations) to support data-driven decisions.
- Partnered with cross-functional teams (Finance, Ops, Risk) to translate business requirements into scalable analytical solutions.

Data Engineer

Infosys | Hyderabad, India | Jun 2018 – Dec 2022

- Designed & optimized complex SQL-based ETL workflows improving warehouse performance by **40%**.
- Developed Python-based automation scripts for data cleansing, transformation, and reconciliation, reducing manual effort **50%**.
- Migrated legacy reporting systems to Power BI and Tableau, modernizing BI platforms for 2,000+ users.
- Built scalable data models (star/snowflake schemas) enabling faster analytics across business units.
- Implemented data validation, anomaly detection, and cleansing pipelines increasing reliability by **30%**.
- Supported predictive analytics workflows with feature extraction, dataset preparation, and pipeline optimization.
- Managed scheduling, orchestration, and monitoring of ETL jobs across multiple on-prem & cloud environments.
- Built REST API integrations to ingest partner and ERP data into the enterprise warehouse.
- Worked closely with data science teams to deploy ML features and model outputs into production systems.
- Ensured data governance through metadata management, lineage documentation, and quality scorecards.

Education

Master's in Computer and Information Science — **Southern Arkansas University**

Bachelor's in Information Technology — **Osmania University**